

NETWORK-BASED INTRUSION DETECTION SYSTEM (NIDS) PROJECT

Submitted By

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Certificate

This is to certify that the project titled “Implementation of a Network-Based Intrusion Detection System Using Snort/Suricata” has been successfully completed as part of the academic requirements.

The work presented in this report is original and has been carried out under proper guidance.

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Acknowledgement

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Network-Based Intrusion Detection System (NIDS) Project

Title

Implementation of a Network-Based Intrusion Detection System Using Snort/Suricata

1. Objective

The objective of this project is to:

- Set up a Network-Based Intrusion Detection System (NIDS)
 - Configure detection rules and alerts
 - Monitor network traffic for malicious activity
 - Implement response mechanisms for attacks
 - Visualize detected threats using dashboards and logs
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2. Introduction

A Network-Based Intrusion Detection System (NIDS) is a cybersecurity solution that monitors and analyzes network traffic to detect suspicious or malicious activities.

Two widely used NIDS tools are:

- Snort
- Suricata

These tools inspect packets in real time and generate alerts whenever suspicious activity is detected.

In this project, Suricata is used because of its:

- High performance
 - Multi-threading support
 - Easy integration with visualization tools
 - Advanced protocol detection capabilities
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3. System Requirements

Hardware Requirements

- Minimum 4 GB RAM
- Dual-core processor
- 20 GB free storage

Software Requirements

- Ubuntu 22.04 / Kali Linux
- Suricata IDS

- *Elasticsearch*
 - *Kibana*
 - *Filebeat*
 - *Wireshark (optional)*
-

4. Network Architecture

Components

1. *Attacker Machine*
2. *Victim Machine*
3. *IDS Monitoring System*
4. *Switch/Router*

Workflow

- *Network packets travel through the network.*
 - *Suricata captures packets.*
 - *Rules analyze traffic.*
 - *Alerts are generated for suspicious activity.*
 - *Logs are stored.*
 - *Kibana dashboard visualizes attacks.*
-

5. Installation and Configuration

Step 1: Update System

`sudo apt update && sudo apt upgrade -y`

Step 2: Install Suricata

```
sudo apt install suricata -y
```

Check installation:

```
suricata --build-info
```

Step 3: Identify Network Interface

```
ip a
```

Example Interface:

```
eth0
```

Step 4: Configure Suricata

Open configuration file:

```
sudo nano /etc/suricata/suricata.yaml
```

Modify interface:

```
af-packet:  
  - interface: eth0
```

Save and exit.

Step 5: Test Configuration

```
sudo suricata -T -c /etc/suricata/suricata.yaml
```

Expected Output:

Configuration provided was successfully loaded.

Step 6: Start Suricata Service

```
sudo systemctl start suricata  
sudo systemctl enable suricata
```

Check status:

```
sudo systemctl status suricata
```

6. Rule Configuration

Suricata rules detect suspicious traffic.

Rule file location:

```
/etc/suricata/rules/local.rules
```

Open rules file:

```
sudo nano /etc/suricata/rules/local.rules
```

Example Rules

1. ICMP Ping Detection

```
alert icmp any any -> any any (msg:"ICMP Ping Detected"; sid:1000001; rev:1;)
```

Purpose: Detects ping scans.

2. SSH Brute Force Detection

```
alert tcp any any -> any 22 (msg:"Possible SSH Brute Force"; threshold:  
type both, track by_src, count 5, seconds 60; sid:1000002; rev:1;)
```

Purpose: Detects multiple SSH attempts.

3. Nmap Scan Detection

```
alert tcp any any -> any any (msg:"Nmap Scan Detected"; flags:S; thresh  
old:type both, track by_src, count 20, seconds 10; sid:1000003; rev:1;)
```

Purpose: Detects reconnaissance scans.

4. HTTP SQL Injection Detection

```
alert http any any -> any any (msg:"SQL Injection Attempt"; content:"un  
ion select"; nocase; sid:1000004; rev:1;)
```

Purpose: Detects SQL injection attempts.

Reload Rules

```
sudo systemctl restart suricata
```

7. Traffic Monitoring

Live Traffic Capture

```
sudo tail -f /var/log/suricata/fast.log
```

OR

```
sudo tail -f /var/log/suricata/eve.json
```

Generate Test Traffic

Ping Test

```
ping 8.8.8.8
```

Port Scan Test

```
nmap -sS <target-ip>
```

SSH Brute Force Simulation

```
hydra -l root -P passwords.txt ssh://<target-ip>
```

8. Response Mechanisms

Intrusion response mechanisms automatically react to attacks.

Method 1: Block IP Using IPTables

```
sudo iptables -A INPUT -s <attacker-ip> -j DROP
```

Purpose: Blocks malicious IP addresses.

Method 2: Automatic Blocking with Fail2Ban

Install Fail2Ban:

```
sudo apt install fail2ban -y
```

Configure jail:

```
sudo nano /etc/fail2ban/jail.local
```

Example:

```
[sshd]  
enabled = true  
maxretry = 5  
bantime = 3600
```

Restart service:

```
sudo systemctl restart fail2ban
```

Method 3: Email Alerts

Install mail utility:

```
sudo apt install mailutils -y
```

Send alert script:

```
echo "Intrusion Detected" | mail -s "Security Alert" admin@example.com
```

9. Visualization Dashboard

Visualization helps security analysts monitor attacks efficiently.

Tools Used:

- *Elasticsearch*
 - *Kibana*
 - *Filebeat*
-

Install Elasticsearch

```
sudo apt install elasticsearch -y
```

Enable service:

```
sudo systemctl enable elasticsearch  
sudo systemctl start elasticsearch
```

Install Kibana

```
sudo apt install kibana -y
```

Enable service:

```
sudo systemctl enable kibana  
sudo systemctl start kibana
```

Install Filebeat

```
sudo apt install filebeat -y
```

Enable Suricata module:

```
sudo filebeat modules enable suricata
```

Setup Filebeat:

```
sudo filebeat setup
```

Start Filebeat:

```
sudo systemctl start filebeat
```

Access Kibana Dashboard

Open browser:

`http://localhost:5601`

Dashboard Features:

- Real-time attack monitoring
 - Top attacker IPs
 - Alert severity graphs
 - Protocol analysis
 - Attack timeline visualization
-

10. Testing and Validation

Test Case 1: Ping Detection

Action:

`ping <target-ip>`

Expected Result:

ICMP Ping Detected

Status: Passed

Test Case 2: Nmap Scan Detection

Action:

`nmap -sS <target-ip>`

Expected Result:

Nmap Scan Detected

Status: Passed

Test Case 3: SQL Injection Detection

Action:

Attempt SQL payload in web application.

Expected Result:

SQL Injection Attempt

Status: Passed

11. Logs and Output

Fast Log Example

05/07/2026-18:42:10 [**] [1:1000001:1] ICMP Ping Detected [**]

EVE JSON Example

```
{
  "timestamp": "2026-05-07T18:42:10.123456+0000",
  "event_type": "alert",
  "src_ip": "192.168.1.10",
  "dest_ip": "192.168.1.20",
  "alert": {
    "signature": "ICMP Ping Detected"
  }
}
```

12. Advantages of the System

- *Real-time monitoring*
 - *Detects multiple attack types*
 - *Easy rule customization*
 - *Scalable architecture*
 - *Supports automated response*
 - *Centralized logging and visualization*
-

13. Limitations

- *False positives may occur*
 - *Requires regular rule updates*
 - *High traffic environments need stronger hardware*
 - *Encrypted traffic is difficult to inspect completely*
-

14. Security Best Practices

- *Update IDS rules regularly*
 - *Monitor logs daily*
 - *Restrict unnecessary ports*
 - *Use firewall integration*
 - *Enable automatic backups*
 - *Implement strong authentication*
-

15. Troubleshooting

Problem: Suricata Service Not Starting

Solution:

```
sudo suricata -T -c /etc/suricata/suricata.yaml
```

Fix configuration errors.

Problem: No Alerts Generated

Solution:

- Verify rules are loaded
 - Check interface configuration
 - Generate test traffic
 - Restart Suricata
-

Problem: Dashboard Not Loading

Solution:

```
sudo systemctl status elasticsearch  
sudo systemctl status kibana
```

Ensure services are running.

16. Conclusion

This project successfully implemented a Network-Based Intrusion Detection System using Suricata.

The system:

- *Monitors live network traffic*
- *Detects malicious activities*
- *Generates real-time alerts*
- *Responds to attacks*
- *Visualizes incidents through dashboards*

The implementation demonstrates how IDS solutions improve organizational cybersecurity by identifying and mitigating threats efficiently.

17. Future Enhancements

- *AI-based anomaly detection*
 - *Cloud IDS deployment*
 - *Machine learning integration*
 - *Automated threat intelligence feeds*
 - *SIEM integration*
 - *Advanced incident response automation*
-

18. References

1. *Suricata Official Documentation*
2. *Snort Official Documentation*
3. *Elasticsearch Documentation*
4. *Kibana Documentation*
5. *Cybersecurity and IDS Research Papers*

19. Final Result

The Network-Based Intrusion Detection System was successfully configured and tested.

All major attack simulations generated alerts successfully.

The dashboard displayed real-time intrusion information accurately.

The project achieved:

✓ Real-time monitoring ✓ Threat detection ✓ Alert generation ✓
Attack visualization ✓ Response implementation ✓ Successful testing

PROJECT STATUS: COMPLETED SUCCESSFULLY